

Reader Report: Week 06

Probabilistic Invariance: VI-BIP and the Core Theorems

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Summary

Wu and Blei (2025) recast ICP as sparse Bayesian variable selection. A binary selector $z \in \{0,1\}^p$ indexes the invariant features; the posterior is

$$p(z \mid \mathcal{D}) \propto p(z) \prod_{e,i} \frac{\hat{g}(Y_i^e \mid X_{ei}^z)}{\hat{p}_e(Y_i^e \mid X_{ei}^z)},$$

where $\hat{g}$ is the pooled conditional and $\hat{p}_e$ is the per-environment conditional. The ratio up-weights selectors z for which the pooled fit matches each local fit—a continuous, probabilistic version of Peters et al.’s accept/reject test for invariance.

Two theorems characterize the posterior’s behavior. **Consistency (Thm 1)**: if $\mu_{\min} > 0$ (positive KL-divergence between pooled and local conditionals on at least one non-invariant feature), the posterior concentrates on the true selector z^* as $n, E \rightarrow \infty$. **Contraction (Thm 2)**: the posterior mass on $\{z \neq z^*\}$ shrinks as $O(R e^{-\kappa n E \mu_{\min}})$, where $R = 2^p$ is the hypothesis count, E the number of environments, n the per-environment sample size, and κ a universal constant. Three practical implications follow. First, environments and data are interchangeable in the exponent: fewer environments can be compensated by more data and vice versa, which is encouraging given that my NIMBYism application has only $K \approx 4$ regimes. Second, $R = 2^p$ appears only in the polynomial prefactor, not the exponent, so the curse of dimensionality is milder than it might appear for large p . Third, $\mu_{\min}$ is a KL quantity measuring how much the non-invariant features actually shift across environments; weakly heterogeneous environments slow contraction regardless of n or E .

VI-BIP approximates $p(z \mid \mathcal{D})$ with a mean-field family $q(z) = \prod_j \text{Bernoulli}(z_j; \pi_j)$, maximizing the ELBO. The key computational challenge is that the ELBO gradient requires differentiating through a discrete draw. The paper applies the **U2G estimator**, which relaxes the discrete $z \sim q$ to a continuous Gaussian surrogate before backpropagating, yielding low-variance, unbiased gradient estimates compatible with standard stochastic optimizers. On a 6,170-gene perturbation dataset, VI-BIP recovers the same invariant gene sets as exact BIP while running orders of magnitude faster.

Critique & Contrast with the Optimization Approach

Fan et al. (2024) offer a useful contrast. They replace the combinatorial posterior with a minimax objective

$$\min_b \max_{w \in U(\gamma)} \sum_e w_e \hat{\mathbb{E}}[(Y^e - b^\top X^e)^2],$$

where $U(\gamma)$ permits negative environment weights to enforce risk parity. NegDRO returns a point estimate $\hat{\beta}^\gamma$ with explicit polynomial-time convergence and finite-sample rates, but no uncertainty quantification. The key conceptual difference is that BIP produces a *distribution* over invariant sets, revealing which features are ambiguous when environments are few, while NegDRO produces a single answer with provable rates but no sense of the remaining ambiguity. For my project with ~ 4 environments and ~ 50 features, the multi-modal structure of the BIP posterior is precisely what I need: it tells me *where* more environments would reduce ambiguity, not just what the current best estimate is.

Questions

1. The contraction rate depends on $\mu_{\min}$ across environments. In practice, one can estimate $\hat{\mu}_e$ from data, but the bound uses the population quantity. Is there a data-driven way to certify that $\mu_{\min}$ is bounded away from zero before fitting BIP, analogous to the pre-test in classical ICP where one first checks whether any invariant set exists?
2. VI-BIP's mean-field approximation factorizes across features, which ignores posterior correlations between z_j and z_k for correlated covariates. In the NIMBYism setting, regulatory features (zoning category, parking requirements) are strongly correlated. Does the mean-field gap systematically bias π_j toward 0.5 for correlated features, and if so, would a structured (e.g. Ising-model) variational family recover sharper posteriors?

References

- Y. Fan, S. Li, and Z. Yang. Causal invariance learning via efficient nonconvex optimization. *arXiv preprint arXiv:2412.11850*, 2024. URL <https://arxiv.org/abs/2412.11850>.
- Y. Wu and D. M. Blei. Bayesian invariance modeling of multi-environment data. *arXiv preprint arXiv:2506.22675*, 2025. URL <https://arxiv.org/abs/2506.22675>.